Hung-Yu Lin

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SUMMARY

Software Engineer with 3+ years of hands-on experience in iOS and Flutter mobile application development, specializing in real-time IoT systems, on-device machine learning, and ARKit-based computer vision pipelines. Experienced in building production mobile applications with BLE, GPS, sensor streaming, AWS IoT integration, and CI/CD automation across 8+ mobile apps. Published ASME researcher with a strong ML background in PyTorch, ONNX Runtime, CoreML, and Physics-Informed Neural Networks, bridging mobile engineering, edge AI inference, and applied machine learning for real-world systems.

SKILLS

Mobile Development: iOS · Flutter · ARKit · CoreML · Mobile UI/UX · App Deployment
Edge AI: On-device ML · ONNX Runtime · Object Detection · AR-based Computer Vision · Real-time Inference
Machine Learning: Physics-Informed Neural Networks · Scientific ML · Deep Learning · Semantic Segmentation
Embedded & IoT: BLE · GPS · Sensor Fusion · AWS IoT · Real-time Telemetry
Cloud, DevOps & Backend Integration: AWS · Cloud Services · Docker · CI/CD · RESTful API Integration · Git

WORK EXPERIENCE

Texas A&M University-Corpus Christi — Texas, USA
Graduate Research Assistant (GRA) — Jan 2025 – Present

- Accelerated geothermal candidate screening for large-scale project pipelines by developing a Physics-Informed Neural Network (PINN) surrogate model in PyTorch that achieved stable thermal predictions under sparse and noisy field data — outperforming finite-element baselines in low-data regimes.
- Enabled data-driven investment decision-making by integrating techno-economic metrics (NPV, LCOE, DPP) into an end-to-end ML evaluation framework for closed-loop geothermal systems, reducing per-site analysis time.
- Published first-author paper accepted to ASME ES2026 (Paper ID: ES2026-184639) presenting a hybrid ORC power estimation and PINN surrogate modeling framework for abandoned oil well repurposing.

MICROPROGRAM INFORMATION CO., LTD — Taichung, Taiwan
Application Engineer — iOS & Flutter — Jan 2024 – Aug 2024

- Reduced bike theft by 30% across a 120K+ daily-user IoT platform by integrating AWS IoT cloud monitoring with real-time BLE telemetry streaming between mobile clients and embedded bike meters.
- Achieved 95% user satisfaction rating by engineering real-time GPS and sensor data pipelines supporting reservations, rentals, and anti-theft queries with fault-tolerant cloud architecture on AWS.
- Improved system latency and reliability for concurrent user loads by collaborating cross-functionally to optimize data flow and fault tolerance across IoT and cloud service layers.

CHUNG YO INTERNET INFORMATION CO., LTD — Taichung, Taiwan
Application Engineer — iOS & Flutter — Oct 2021 – Jun 2023

- Cut deployment time by 26% and production bugs by 50% by building Jenkins and AWS CI/CD pipelines with automated testing across 8+ mobile applications.
- Scaled real-time data streaming platforms to support concurrent user peaks by profiling performance bottlenecks, optimizing processing pipelines, and implementing systematic monitoring.

EDUCATION

Texas A&M University-Corpus Christi Texas, USA
Master of Science in Computer Science Aug 2024 – Expected Aug 2026

- GPA: 3.5 / 4.0 · Graduate Research Assistant · PI: Dr. Chen, Associate Dean in College of Engineering
- Coursework: Deep Learning, Machine Learning, Data Structures & Algorithms

National Kaohsiung University of Science and Technology Taichung, Taiwan
Bachelor of Science in Computer Science and Information Engineering Aug 2016 – Jun 2020

- GPA: 3.6 / 4.0

PUBLICATION

Lin, H.-Y., Shih, K.-C., & Chen, L.-D. **Techno-Economic Analysis of Repurposing Abandoned Oil Wells for Geothermal Energy Extraction by Physics-Informed Neural Network**. ASME Energy Sustainability Conference (ES2026), Bellevue, WA. Paper ID: ES2026-184639.

- Developed a PINN surrogate model for stable geothermal thermal prediction under sparse and noisy data conditions, integrated with ORC-based power estimation and NPV/LCOE/DPP economic metrics for system-level feasibility screening.

PROJECTS

AR Navigation System: iOS · ARKit · CoreML · ONNX Runtime · A* Algorithm GitHub Link

- Built an on-device edge ML system for object localization and recall, running real-time object detection fully on-device via CoreML and ONNX Runtime with no backend dependency, achieving zero-latency inference.
- Designed a persistent object marking system that stores labeled landmarks in a spatial map, enabling users to save, retrieve, and navigate back to previously detected objects across sessions.
- Applied A* pathfinding over AR mesh-reconstructed maps and implemented dynamic route updates to maintain stable navigation as the environment changes.

Local Memory AI: RAG · LLM · Fast API ·Privacy-Oriented AI GitHub Link

- Built a privacy-preserving local AI knowledge vault for macOS that ingests PDFs, images, and text files, enabling natural language Q&A over personal documents with full source citations.
- Designed a Guardian Pipeline with a local LLM (Ollama) to classify query sensitivity and automatically redact PII before routing to cloud models, ensuring sensitive data never leaves the device.
- Implemented semantic search using BGE-M3 multilingual embeddings with sqlite-vec for on-device ANN retrieval, achieving low-latency responses without external vector database dependencies.